

Lab 3: Component Modelling & Architectural Pattern Selection

Student Club Event Ticketing & Budget Portal

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1. Scenario Review

The Student Club Event Ticketing & Budget Portal supports event proposal submission, multi-stage approval, proposal tracking, QR-code ticket generation and venue validation. The approval workflow proceeds through the Faculty Coordinator, Finance Officer and Dean. The system also requires authenticated role-based access, reliable concurrent data handling, and fast QR validation, with at least 95% of validation requests expected to complete within 100 ms under normal conditions.

2. Architectural Style Analysis

Architecture	Suitability for this system
Layered	Strong fit. The portal can be separated into presentation, business logic and data layers, giving clear separation of responsibilities and straightforward maintenance.
Microservices	Supports independent scaling and fault isolation, but adds network, deployment and data-consistency complexity that is unnecessary for the current portal scope.
Client-Server	Simple and centralized, but the server can become a scalability bottleneck and a single point of failure.

3. Architecture Selection

Selected Architecture: Layered Architecture

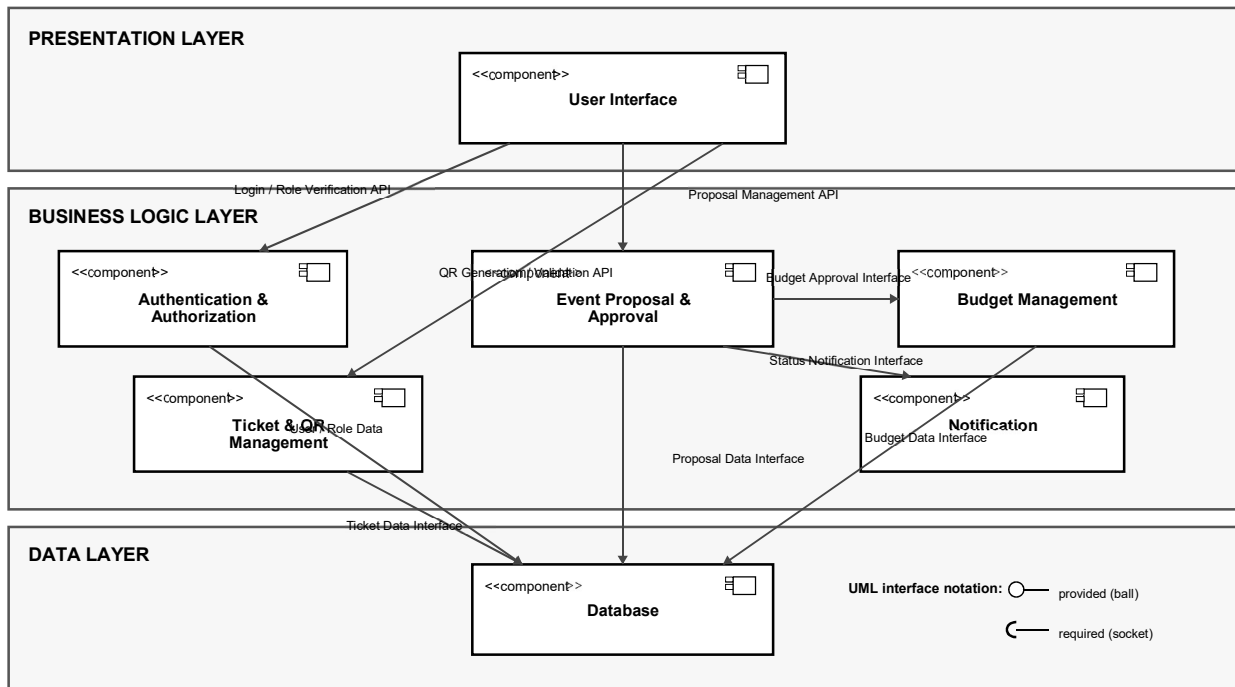
Layered Architecture is selected because the portal naturally separates into a user-facing presentation layer, a business layer containing authentication, approvals, budget handling, ticketing and notifications, and a data layer for persistent storage. This structure is easy to understand and matches the modular nature of the system.

4. Identified Components

Component	Responsibility
User Interface	Forms, dashboards, proposal tracking, approval screens and QR interactions.
Authentication & Authorization	Login, identity checks and role-based access control.
Event Proposal & Approval	Proposal creation, status changes and Faculty -> Finance -> Dean workflow.
Budget Management	Budget review and approval functions.
Ticket & QR Management	Unique QR ticket generation, scanning and validation.
Notification	Approval and rejection status notifications.
Database	Stores users, roles, proposals, approvals, budgets, tickets and check-in data.

5. UML Component Diagram

The diagram below shows the selected layered structure, the major system components, and labelled interfaces/dependencies between them.



6. Written Justification

Architecture Selection. We chose **Layered Architecture** for the Student Club Event Ticketing & Budget Portal.

Reason 1 - Separation of concerns. The system contains distinct responsibilities including authentication, event proposal approval, budget management, QR ticket management, notifications and data storage. A layered structure keeps the user interface separate from business rules and persistent data, making the design easier to understand and maintain.

Reason 2 - Maintainability and controlled dependencies. Changes can be localized to the relevant layer or component. For example, QR ticket handling can be modified without redesigning the proposal approval workflow, while database access remains isolated in the data layer.

Security Advantage. Authentication and authorization are handled before protected business operations are allowed. This supports the portal requirement that only authenticated and authorized users may access administrative functions and approval workflows.

Performance Benefit. The business layer can keep QR validation logic focused and use direct, well-defined data access interfaces. This design supports the stated performance target that at least 95% of QR-code validation requests return within 100 ms under normal operating conditions.

7. Key Interfaces Shown

Interface	Purpose
Login / Role Verification API	UI requests authentication and role verification.
Proposal Management API	UI creates, views and tracks event proposals.
Budget Approval Interface	Proposal workflow communicates with budget management.
Status Notification Interface	Approval/rejection events trigger notifications.
Proposal Data Interface	Proposal and approval data is stored/retrieved.
Ticket Data Interface	Ticket and check-in data is stored/retrieved.
QR Generation / Validation API	UI/event staff interact with QR ticket services.